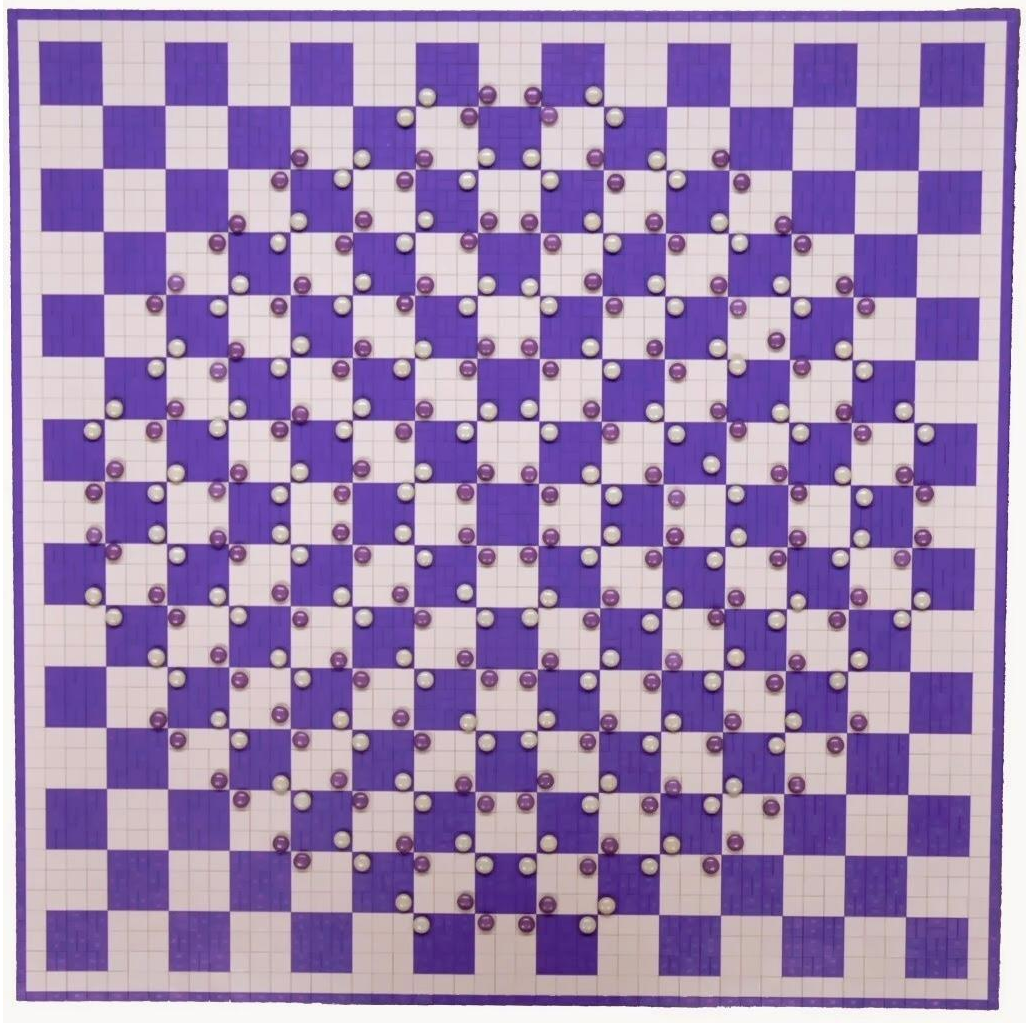
Can computers match human perception?



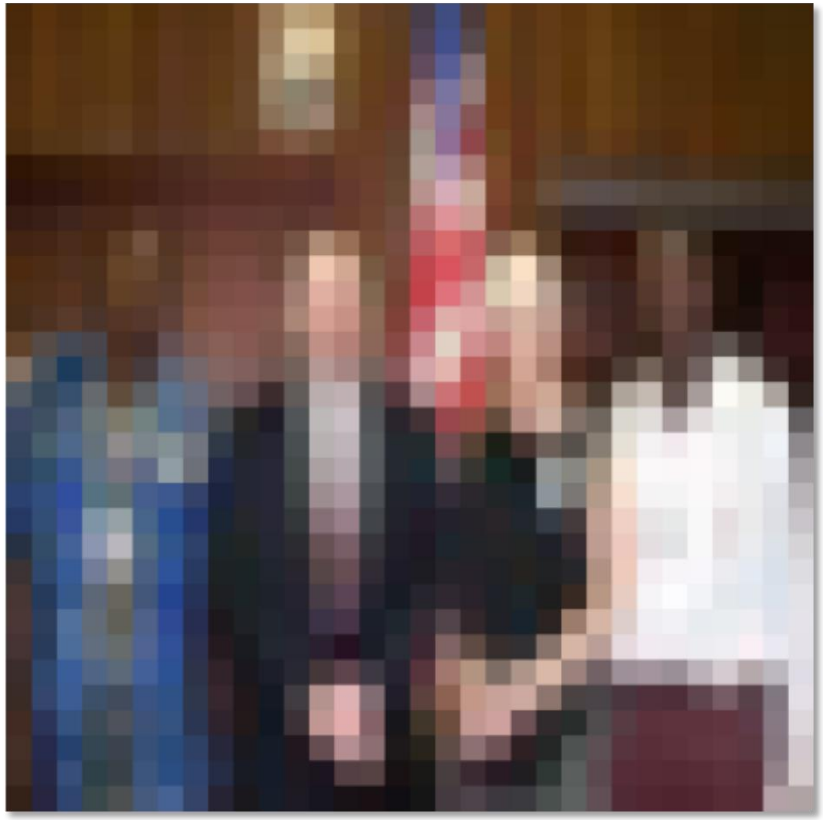
- Yes and no
  - humans are better at “hard” things, are more robust, and make inferences quickly and cheaply
- But huge progress
  - Accelerating in the last 10 years due to deep learning, large vision-language models
  - What is considered “hard” keeps changing

Human perception has its shortcomings



https://twitter.com/pickover/status/1460275132958662657/

But humans can tell a lot about a scene from a little information…



Source: “80 million tiny images” by Torralba, et al.

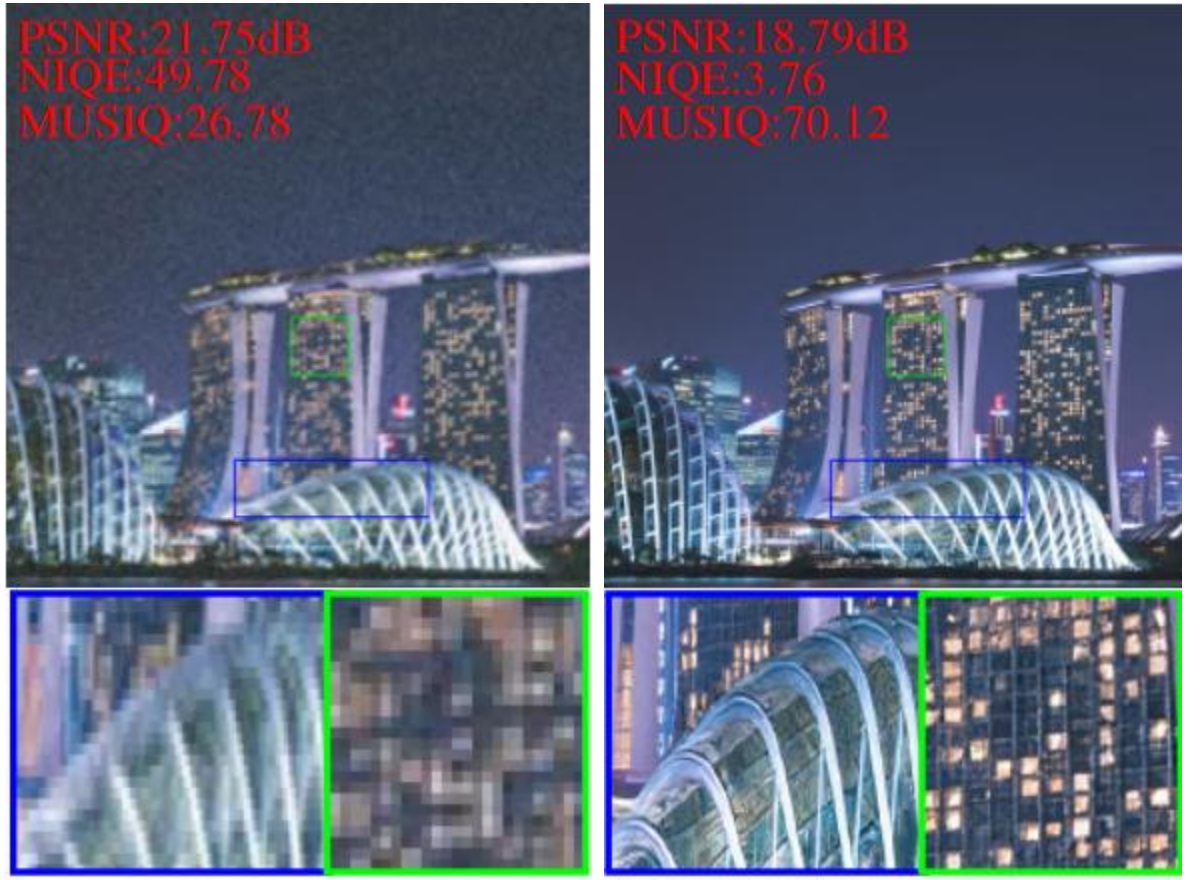


The goal of computer vision



The goal of computer vision

- “Enhance” images



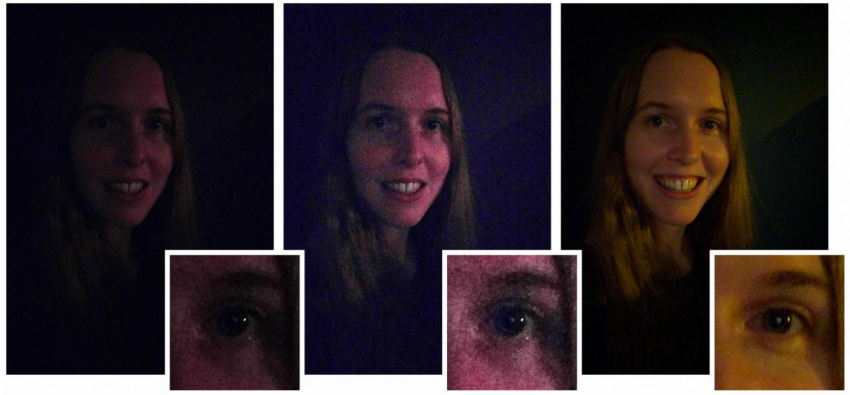


The goal of computer vision

- "Enhance" images



Super-resolution (source: 2d3)



Low-light photography
(credit: Hasinoff et al., SIGGRAPH ASIA 2016)



Depth of field on cell phone camera
(source: Google Research Blog)



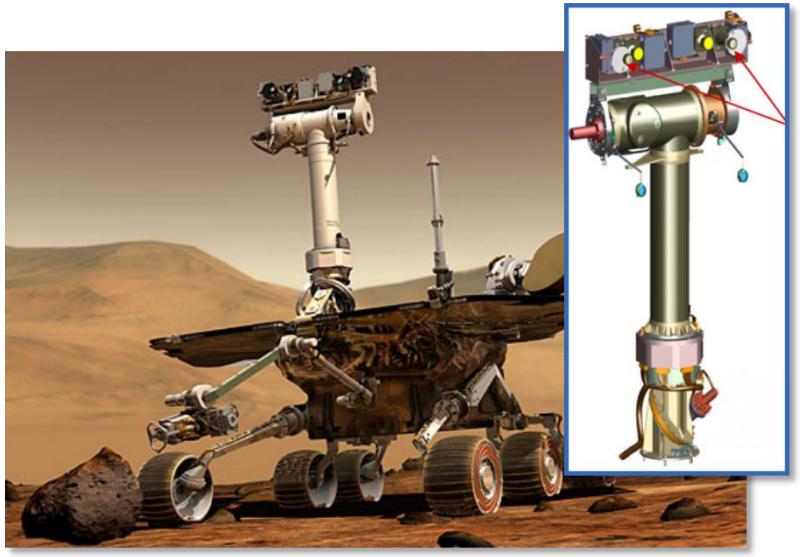
Removing objects
(Google Magic Eraser)

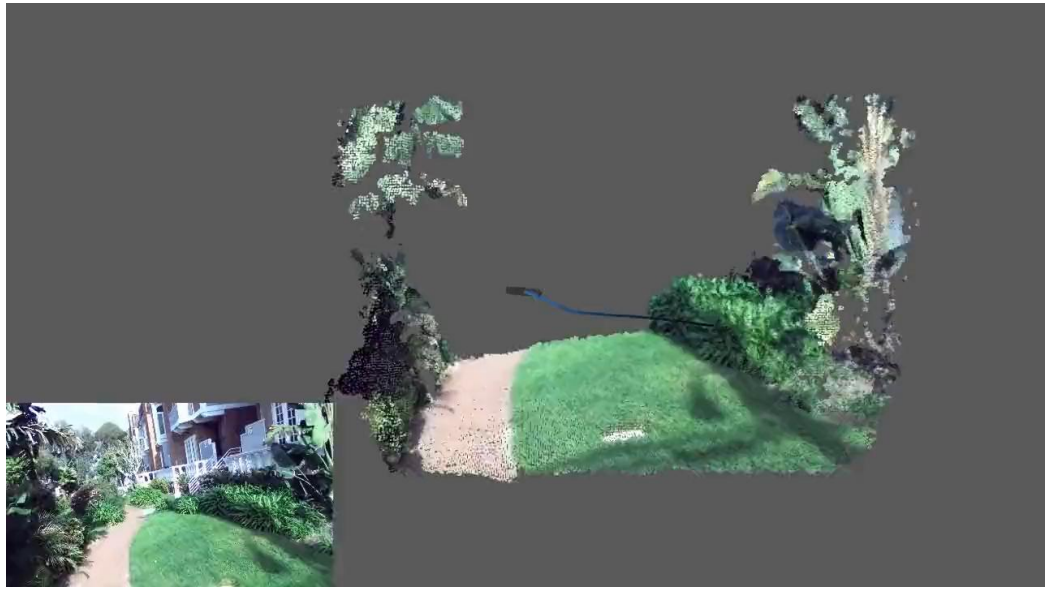
The goal of computer vision

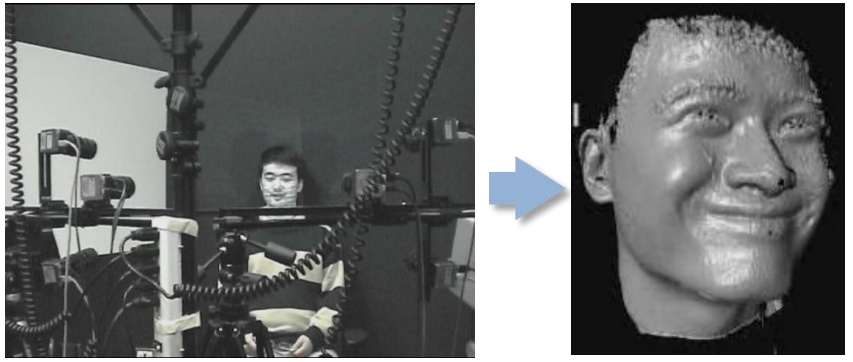
- Compute the 3D shape of the world



ZED 2i Camera

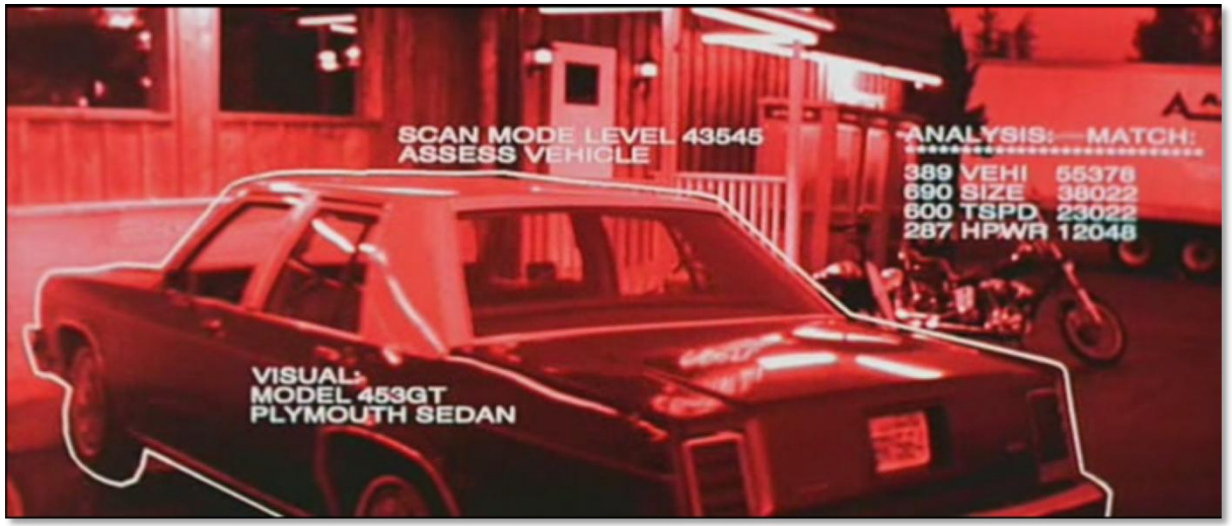






The goal of computer vision

- Recognize objects and people



Terminator 2, 1991